

Origin essentialism

PHIL UA-78 Metaphysics

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Salmon's reconstruction of Kripke's argument from footnote 56

Sufficiency For any table t and hunk h : if it is possible for t to be a table originally formed from h , then it is impossible for a table distinct from t to be originally formed from h .

Origin Uniqueness Necessarily, no table is originally formed from two distinct hunks.

Weak Compossibility For any two non-overlapping hunks, if each is such that it is possible for there to be a table originally formed from it, it is possible that both of them originally form tables.

Necessity of Distinctness For any objects x and y , if $x \neq y$, then it is impossible for it to be the case that $x = y$.

Overlap Essentialism For any non-overlapping hunks h_1 and h_2 and table t , if t is originally formed from h_1 , it is impossible for t to be originally formed from h_2 .

Why is this argument valid?

1. Suppose for *reductio ad absurdum* that t is a table and h_1 and h_2 are non-overlapping hunks, such that t could have been a table originally formed from h_1 and also could have been a table originally formed from h_2
2. By Sufficiency, it is necessary that any table originally formed from h_1 is identical to t , and also necessary that any table originally formed from h_2 is identical to t .

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3. But by Weak Compossibility, it is possible that both h_1 and h_2 are such that some table is originally formed from them.
4. Hence it is possible that t is a table originally formed from both h_1 and h_2 .

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4. Hence it is possible that t is a table originally formed from both h_1 and h_2 .
5. So by Origin Uniqueness, it's possible that $h_1 = h_2$.
6. But this contradicts the Necessity of Distinctness.

The other premises: Origin Uniqueness

Origin Uniqueness seems unproblematic given the intended interpretation of 'originally formed from'.

- Of course we could combine two or more hunks of wood to form a table! But the resulting table is 'originally formed' not from any of those small hunks, but from a bigger hunk that each of them is part of.

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But if we wanted to be cautious, we could avoid the need for this premise by replacing 'non-overlapping' throughout the argument with 'such that it is possible that both are made into distinct tables'. This corresponds to Kripke's parenthetical remark '(We assume that there is no relation between A and C which makes the possibility of making a table from one dependent on the possibility of making a table from the other.)'

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The conclusion will then be:

Weaker Origin Essentialism For any hunks h_1 and h_2 such that it's possible that h_1 and h_2 are both made into two distinct tables, and any table t that is originally formed from h_1 , it is impossible for t to be originally formed from h_2 .

The other premises: Necessity of Distinctness

The Necessity of Distinctness (ND) is interesting and controversial.

The SEP article by Robertson Ishii tries to make it seem unproblematic by talking about 'the necessity of identity/distinctness', suggesting that somehow these are a package deal. But this is very misleading! *The Leibniz's Law argument for NI does not establish ND.*

The other premises: Necessity of Distinctness

Immediately after the footnote 56 passage quoted last time, Kripke suggests two arguments for ND! Here is the first:

- » However, the same types of considerations that can be used to establish the latter can be used to establish the former. (Suppose $X \neq Y$; if X and Y were both identical to some object Z in another possible world, then $X = Z$, $Y = Z$, hence $X = Y$.)

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There must be some blunder here. It looks like the only properties of identity being appealed to are transitivity and symmetry. But there are many transitive and symmetric relations R for which it can be false that x bears R to y without it being necessarily false—e.g. *being the same height as*.

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Kripke seems to be simply assuming that if $X = Z$ at some possible world w , then $X = Z$. But that's just equivalent to ND, which he's supposed to be arguing for!

The other premises: Necessity of Distinctness

Here's Kripke's second argument:

- » Alternatively, the principle follows from the necessity of identity plus the 'Brouwersche' axiom, or, equivalently, symmetry of the accessibility relation between possible worlds.

Here, Kripke is referring to the following principle, named for L.E.J. Brouwer:

- B** If something is the case, then it is necessarily possible for it to be the case.

The other premises: Necessity of Distinctness

FYI: 'The accessibility relation between possible worlds' can be defined as follows:

World w is accessible from world $v := \text{At } v, w \text{ is a } \textit{possible} \text{ world}.$

The symmetry of accessibility entails that the actual world A is a possible world at every possible world, hence that A is necessarily a possible world, hence that everything that is the case is necessarily possibly the case.

(We can also use the *necessity* of B to go the other way, but that's a bit more complicated.)

The other premises: Necessity of Distinctness

B and (the necessity of) NI jointly entail ND:

- Suppose $x \neq y$. Then by B, it is necessarily possible that $x \neq y$ —i.e. it is necessarily not necessary that $x = y$. But by the necessity of NI, it is necessary that if $x = y$ then it is necessary that $x = y$, and hence necessary that if it's *not* necessary that $x = y$, then $x \neq y$. So, it is necessary that $x \neq y$.

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But why believe B??? I don't know any good snappy arguments.

The other premises: Necessity of Distinctness

Fortunately, we can drop ND as a premise if we just slightly weaken the conclusion, as follows:

For any non-overlapping hunks h_1 and h_2 , if t is originally formed from h_1 , it is impossible for t to be originally formed from h_2 *and not from* h_1 .

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If we combine this move with the previous weakening, we get

Weakest Overlap Essentialism For any hunks h_1 and h_2 that could be made into two distinct tables and any table t originally formed from h_1 , it is impossible for t to be originally formed from h_2 and not from h_1 .

This follows from Sufficiency and Origin Uniqueness.

Problems with Sufficiency: plans

Consider a table t made in 1760 by Thomas Chippendale from a certain hunk h . Sufficiency implies that if h had, instead, been stored until 1970 and then made into an austere cubical modernist table by designer Milo Baughman, then t would have been a modernist table. That may seem dubious!



Problems with Sufficiency: plans

Response (Salmon 1979): weaken Sufficiency to:

Plan Sufficiency For any table t , hunk h , and plan P : if it is possible for t to be a table originally formed from h according to P , then it is impossible for a table distinct from t to be originally formed from h according to P .

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The original argument for Overlap Essentialism still goes through if we strengthen Weak Compossibility to:

Weak Plan Compossibility If h_1 and h_2 are non-overlapping hunks, h_1 could be made into a table according to P_1 , and h_2 could be made into a table according to P_2 , then it could be that: h_1 is made into a table according to P_1 and h_2 is made into a table according to P_2 .

(We also need to add the premise that necessarily every table is made according to some plan—this seems unproblematic.)

A single hunk of wood can be shaped into a table; disassembled; and later made into a table again, according to the same or a different plan.

Sufficiency implies that whenever this happens, we get the same table both times. Plan Sufficiency has the same implication in situations where the same plan is used both times. That may seem dubious!

Salmon's argument against Plan Sufficiency

Maybe it's not so bad to think that these are cases where we get the same table again—after all, things do seem to routinely survive disassembly and reassembly.

Salmon (1979) gives an argument against Sufficiency and Plan Sufficiency based on the following cases.

Table Of Theseus With Reassembly A table is made in 1900 out of hunk h according to plan P . Over the next fifty years, bits of h are gradually replaced, until finally by 1950 none of the original bits of h are part of a table. Then, all the bits of h are gathered and reassembled into a table, again according to P .

Salmon's judgment: The original table survives the part-replacements and is in 1950 composed of matter not overlapping h . So, it is not the table made of h in 1950.

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(This is non-obvious! Hobbes 1655 gives essentially the same case with ships, and uses it to argue that no ship can survive replacement of any of its parts.)

Weakening Sufficiency

In response, Salmon further weakened Sufficiency:

Only-table Plan Sufficiency For any table t , hunk h , and plan P : if it is possible for t to be *the only table ever originally formed from h* , and formed according to P , then it is impossible for a table distinct from t to be the only table originally formed from h and formed according to P .

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This of course requires weakening the conclusion:

Only-table Overlap Essentialism If h_1 and h_2 are non-overlapping hunks and t is the only table ever originally formed from h_1 , then it is impossible for t to be the only table ever originally formed from h_2 .

Robertson Ishii (1998) pointed out a problem:

Not Quite Theseus At possible world w , hunk h is shaped into a table at t ; this table is disassembled, and then much later at t' , a hunk h' almost but not entirely overlapping h is shaped into a table according to the same plan. None of this wood is ever again used for tables.

At w' , almost the same story plays out except that h' is used at t and h is used at t' .

Only-table Plan Sufficiency implies that the table made at t at w is made at t' at w' and vice versa. This seems implausible.

Weakening Sufficiency

In response, Salmon yet again weakened Sufficiency:

No-overlap Plan Sufficiency For any table t , hunk h , and plan P : if it is possible for t to be *the only table ever formed from any matter overlapping h* , and formed according to P , then it is impossible for a table distinct from t to be the only table ever formed from any matter overlapping h and formed according to P .

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It would be understandable if all these weakenings shook our confidence that there's anything true in the ballpark!